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B.Tech. Degree VI Semester Special Supplementary Examination

January 2016

ME 601 INSTRUMENTATION AND CONTROL SYSTEMS

(2006 scheme)

Time : 3 Hours

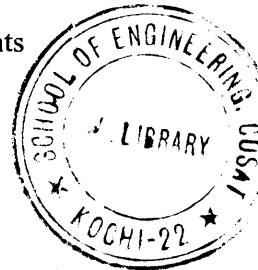
Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Obtain the step response of a first order instrument.
- (b) Explain the difference between active and passive transducer.
- (c) Explain the 5 laws of thermocouple.
- (d) Explain with block diagram the working of a microphone.
- (e) Explain liquid level system with the help of an example.
- (f) Write the analogous electrical elements in Torque-Current analogy for elements of elements of mechanical rotational system.
- (g) State and explain Routh-Hurwitz criterion for determination of stability.
- (h) Explain stepper motor with the help of a diagram.



PART B

(4 × 15 = 60)

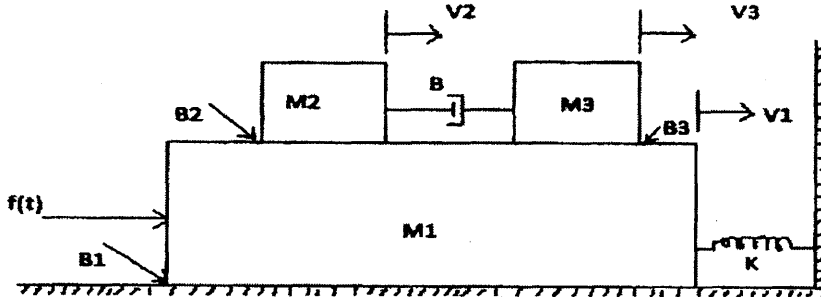
- II. (a) Explain the functional elements of a measuring system with the help of an example. **(10)**
 - (b) Explain second order instrument with the help of an example. **(5)**
- OR**
- III. (a) Explain the methods for correction for interfering and modifying inputs. **(10)**
 - (b) By considering potentiometer transducer explain zero order instruments. **(5)**
 - IV. (a) Explain optical pyrometer with a neat sketch. **(8)**
 - (b) Explain Geiger Muller counter with a neat sketch. **(7)**
- OR**
- V. (a) Explain how hydraulic dynamometer is used to find the torque of the system. **(10)**
 - (b) Explain the difference between bonded and un bonded strain gauge. **(5)**

- VI. The open loop transfer function of a unity feedback control system is given by (15)

$G(S) = \frac{100}{s(s+2)(s+1)}$. For a unit step input, find the time response of a closed loop system and determine percentage over shoot and rise time.

OR

- VII. Consider the mechanical translation system as shown in the figure below. Draw the force-voltage and force current electrical analogous circuits and verify by writing mesh and node equations. (15)



- VIII. Sketch Bode plot for the following open loop transfer function and from the plot determine phase margin and gain margin $G(S) = \frac{30(1+0.1s)}{s(1+0.01s)(1+s)}$. (15)

OR

- IX. The open loop transfer function of a unit feedback system is given as (15)

$G(S) = \frac{k(s+2)}{(s+3)^2(s^2+2s+17)}$. Sketch the root locus of the system.